BASE CHANGE FOR DIRECT IMAGES ON A MORPHISM OF RINGED SIMPLICIAL SITES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the abelian engine is Lemma lemma-direct-image-morphism-simplicial-sites (in-source stub); external tool from Cohomology on Sites.

Editorial note. The label lemma-direct-image-morphism-simplicial-sites-modules names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Abelian case. Same unit, Lemma lemma-direct-image-morphism-simplicial-sites. The isomorphism (g'_n)^{-1}Rh_{total,*}K_ab = Rh_{n,*}g_n^{-1}K_ab proven there is the engine of the reduction.
- Unbounded modules vs abelian sheaves. Cohomology on Sites, Lemma sites-cohomology-lemma-modules-abelian-unbounded. Identifies the derived image of a module complex with the abelian computation, transferring the isomorphism.

Source reference. Original reference 0D97: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1353–1353. Prerequisite pointer invoked at proof line(s) 1677; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DH0: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1604–1604. Prerequisite pointer invoked at proof line(s) 1658; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D72: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1459–1459. Prerequisite pointer invoked at proof line(s) 1665; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0D6J: sources/stacks-chow/tags.txt, source lines 16524–16524. Prerequisite pointer invoked at proof line(s) 1672; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original remark 07A7: sources/stacks-chow/tags.txt, source lines 9297–9297. Prerequisite pointer invoked at proof line(s) 1670; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *With notation and hypotheses as in Lemma 0DH0. For $K \in D(\mathcal{O})$ we have $(g'_n)^* Rh_{total,*} K = Rh_{n,*} g_n^* K$.*

Proof. Recall that $g_n^* = g_n^{-1}$ because $g_n^{-1}\mathcal{O} = \mathcal{O}_n$ by the construction in Lemma 0D72. In particular g_n^* is exact and Lg_n^* is given by applying g_n^* to any representative complex of modules. Similarly for g'_n. There is a canonical base change map $(g'_n)^* Rh_{total,*} K \to Rh_{n,*} g_n^* K$, see Cohomology on Sites, Remark 07A7. By Cohomology on Sites, Lemma 0D6J the image of this in $D(\mathcal{C}'_n)$ is the map $(g'_n)^{-1} Rh_{total,*} K_{ab} \to Rh_{n,*} g_n^{-1} K_{ab}$ where K_{ab} is the image of K in $D(\mathcal{C}_{total})$. This we proved to be an isomorphism in Lemma 0D97 and the result follows. $\square$